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(54) **DISPOSABLE MEDICAL FLOW-REGULATING DEVICE AND SYSTEM**

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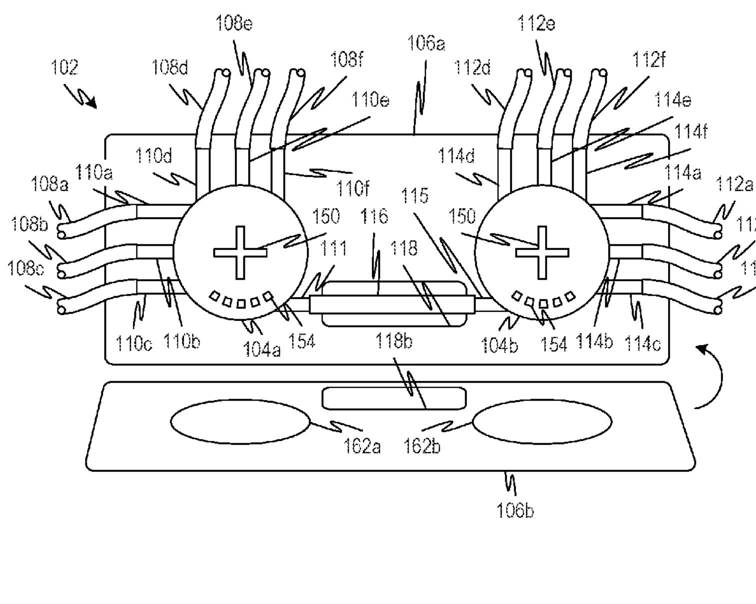
(57) **ABSTRACT**

A disposable medical flow-regulating assembly includes flow-directing units, with multiple fluid-flow lines entering each of the flow-directing units. The flow-directing units are interconnected by a fluid-flow line that extends between them. Each of the flow-directing units includes a rotational insert member that regulates which of multiple flow passages through the flow-directing unit is open and which are closed, based on the angular position of the insert member.

7 Claims, 3 Drawing Sheets



(51) **Int. Cl.**

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F04B 43/00	(2006.01)
F04B 43/12	(2006.01)
F16K 11/085	(2006.01)
F16K 37/00	(2006.01)

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See application file for complete search history.

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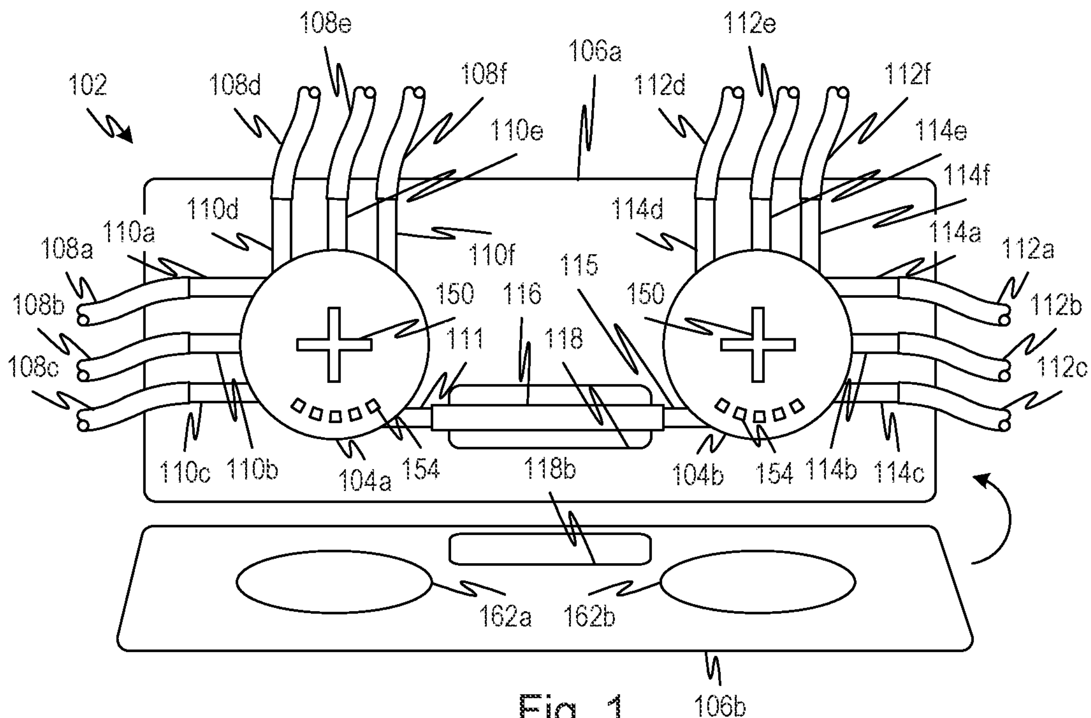
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114a
114b
114c
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118b
150
154
104a
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162a
162b
106b

Fig. 1

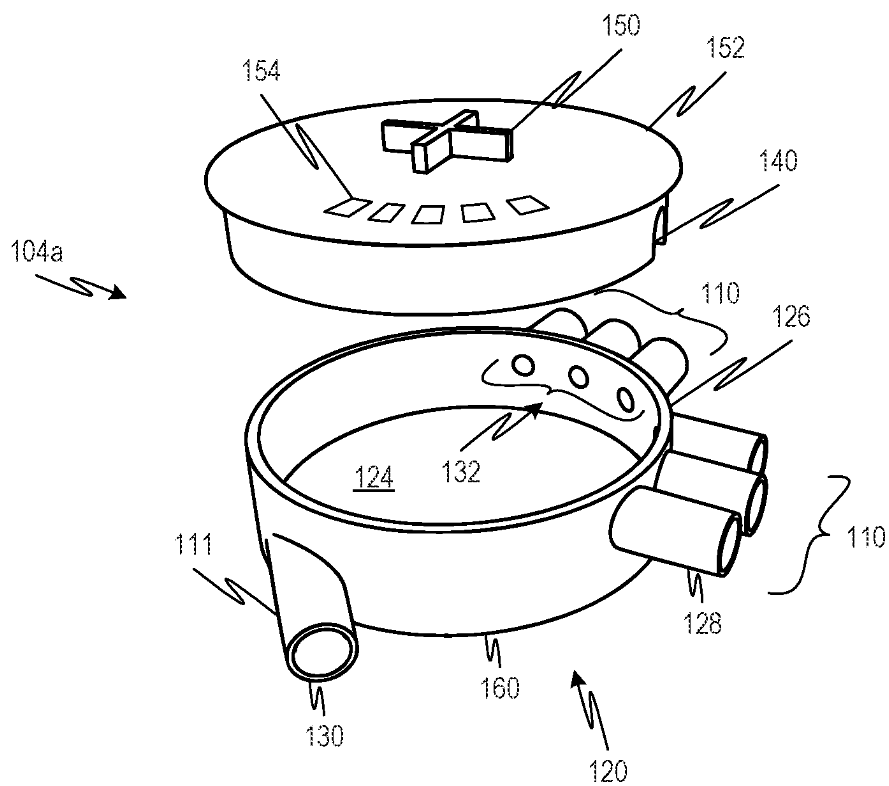
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Fig. 2A

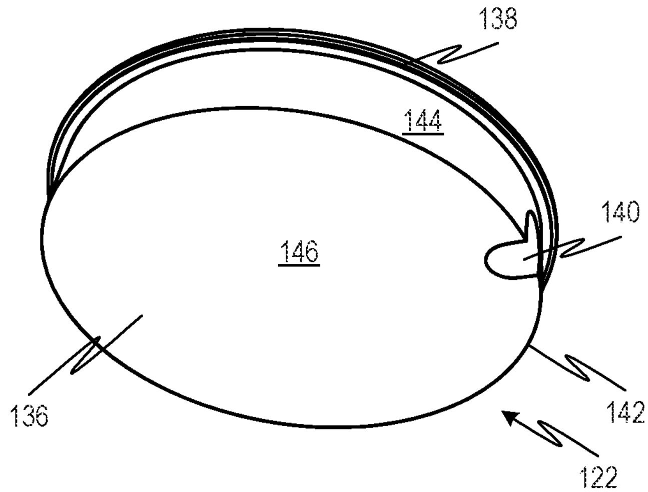
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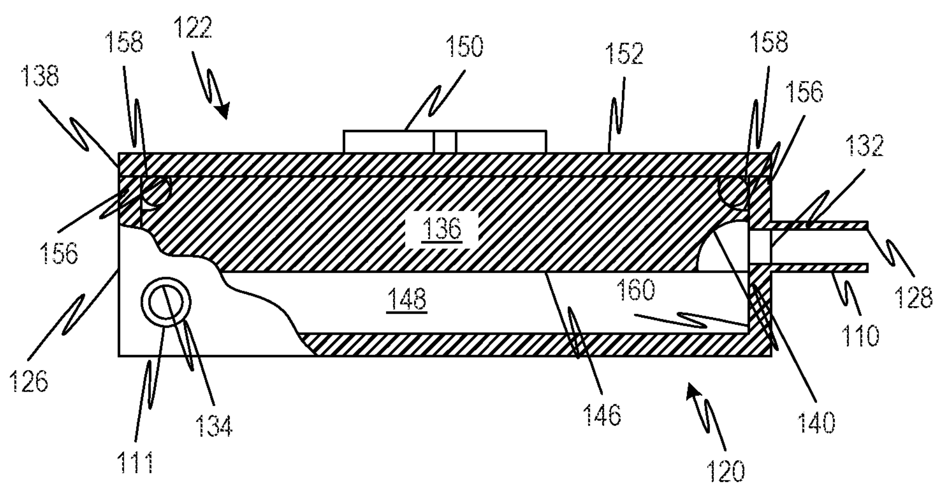
Fig. 2B


122
158
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148
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128
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126
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111
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140
120

Fig. 2C

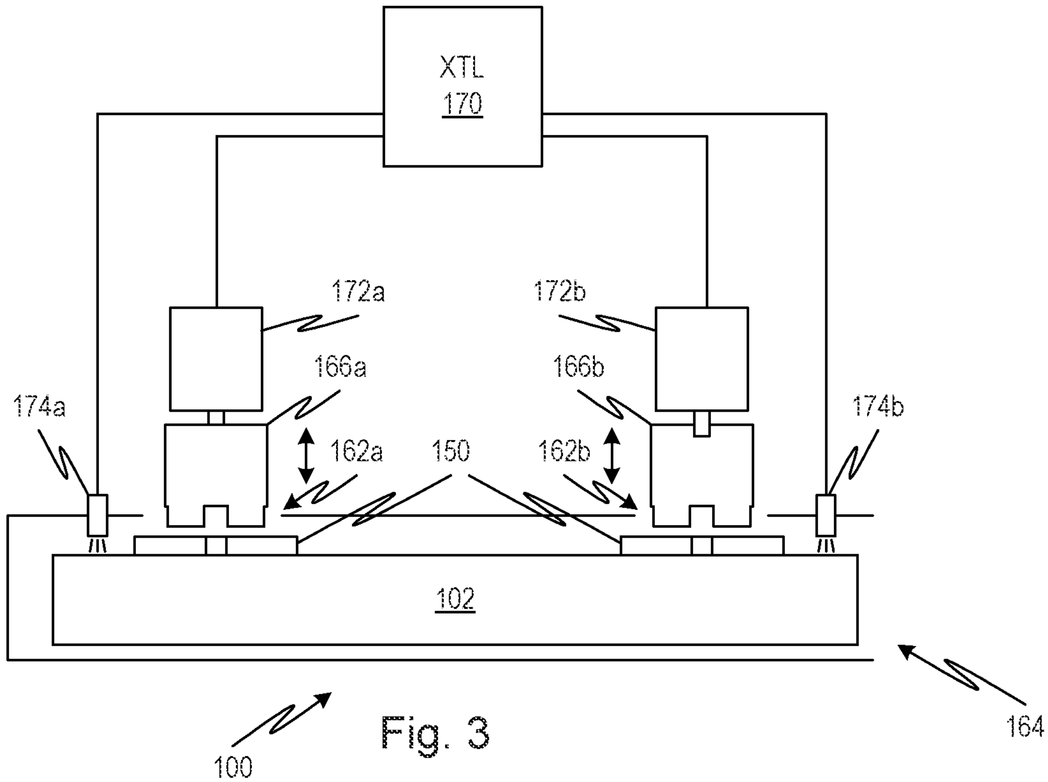

Fig. 3

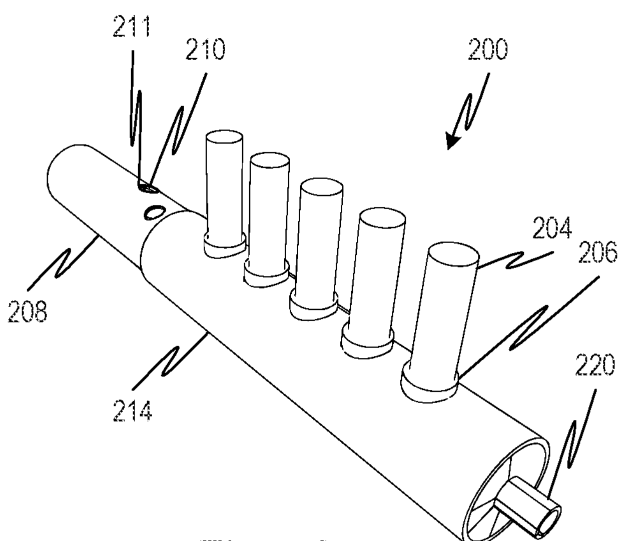

Fig. 4A

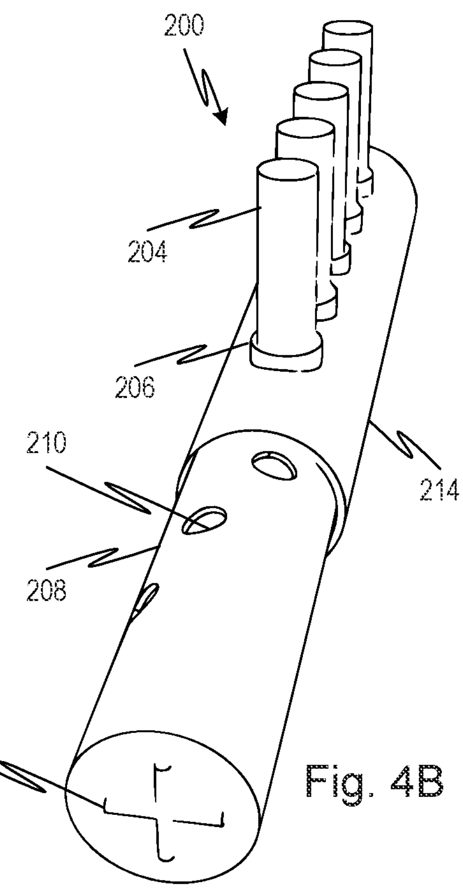

Fig. 4B

DISPOSABLE MEDICAL FLOW-REGULATING DEVICE AND SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This Application is a U.S. national stage filing under 35 U.S.C. § 371 of International Application No. PCT/US2020/027246, filed Apr. 8, 2020, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/831,310 filed Apr. 9, 2019, both of which is are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

This application is directed to a flow-regulating arrangement, particularly suited for use in an automated peritoneal dialysis system.

BACKGROUND

In an automated peritoneal dialysis system as disclosed, for example, in U.S. Pat. No. 9,861,733, incorporated herein by reference, there are a pair (at least) of fluid distribution manifolds. Each of the manifolds has an elongated configuration and two or more ports that permit fluid to flow into or out of a central chamber that runs the length of the manifold, and tubing through which various treatment fluids flow is connected to the various ports. One length of tubing extends from an outlet port of one manifold to an inlet port of the other manifold, passing through a peristaltic pump that drives fluid between the manifolds and hence throughout the various fluid-flow lines of tubing. This is illustrated, for example, in FIGS. 7A-7D and FIGS. 8A and 8D of U.S. Pat. No. 9,861,733 and the accompanying text.

Furthermore, U.S. Pat. No. 9,861,733 illustrates in FIG. 8D an arrangement in which the manifolds and peristaltic pump are housed within a clamshell-type cassette case, and the various lines of tubing leading to and from the manifolds pass across openings formed in the halves of the cassette case. As further explained in the patent, flow through the various lines of tubing may be permitted or prevented by means of a linear actuator (solenoid clamp, stepper and screw drive, pinching mechanism like a plier grip, or other kind of mechanism) that is able to access and selectively clamp each of the various lines of tubing through its respective opening.

Because the fluid distribution assembly is to be used for medical treatment, it is important that flow through the various lines of tubing be controlled (i.e., permitted or prevented) with assurance. Additionally, as a medical device that is intended to be disposed of, cost of manufacture is a consideration.

SUMMARY OF THE INVENTION

A disposable medical flow-regulating assembly includes flow-directing units, with multiple fluid-flow lines entering each of the flow-directing units. The flow-directing units are interconnected by a connecting fluid-flow line that extends between them. The fluid connecting flow line may be a pumping tube segment that is adapted to form a peristaltic pump in cooperation with a roller actuator. Each of the flow-directing units includes a rotational insert member that regulates which of multiple flow passages through the flow-directing unit is open and which are closed, based on the angular position of the insert member.

Objects and advantages of embodiments of the disclosed subject matter will become apparent from the following description when considered in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will hereinafter be described in detail below with reference to the accompanying drawings, wherein like reference numerals represent like elements. The accompanying drawings have not necessarily been drawn to scale. Where applicable, some features may not be illustrated to assist in the description of underlying features.

FIG. **1** is a plan view illustrating a disposable porting cassette in accordance with the disclosed subject matter according to embodiments of the disclosed subject matter.

FIG. **2**A is an assembly view illustrating a flow-directing unit used with the porting cassette illustrated in FIG. **1** according to embodiments of the disclosed subject matter.

FIG. **2**B is a bottom perspective view of the insert member shown in FIG. **2**A according to embodiments of the disclosed subject matter.

FIG. **2**C is a side view, partially in section, of the flow-directing unit illustrated in FIG. **1** according to embodiments of the disclosed subject matter.

FIG. **3** is a schematic diagram illustrating the use of the porting cassette shown in FIG. **1** in the flow-porting section of an automated peritoneal dialysis system according to embodiments of the disclosed subject matter.

FIGS. **4**A and **4**B show respective aspects of a longitudinal cylindrical flow switch configuration according to embodiments of the disclosed subject matter.

DETAILED DESCRIPTION

A flow-regulating system **100** in accordance with the disclosed subject matter is illustrated in FIGS. **1**, **2**A-2C, and **3**. The system utilizes a disposable porting cassette **102**, which includes a pair of disc-shaped, rotary flow-directing units **104***a*, **104***b* that are housed within a clamshell-type cassette casing having side parts **106***a* and **106***b*.

As shown in FIG. **1**, multiple incoming-fluid delivery tubes **108***a*-**108***f*, made from medical-grade material, are attached to respective tube-attachment fittings **110***a*-**110***f* projecting from the flow-directing unit **104***a*, and multiple outgoing-fluid delivery tubes **112***a*-**112***f*, also made from medical-grade material, are attached to respective tube-attachment fittings **114***a*-**114***f* projecting from the flow-directing unit **104***b*. Additionally, a pumping tube segment **116** extends between the flow-directing units **104***a* and **104***b*, attached respectively to tube-attachment fitting **111** projecting from the flow-directing unit **104***a* and tube-attachment fitting **115** projecting from the flow-directing unit **104***b*. The pumping tube segment **116** passes through a peristaltic pump **118**, which can drive fluid in either direction through the pumping tube segment **116** depending on the direction of rotation of the peristaltic pump's actuator (not shown). The incoming-fluid delivery tubes **108***a*-**108***f* may be attached to the tube-attachment fittings **110***a*-**110***f* and outgoing-fluid delivery tubes **112***a*-**112***f* may be attached to respective tube-attachment fittings **114***a*-**114***f* by bonding or other suitable means known in the art.

Construction of the rotary flow-directing unit **104***a* (e.g., with tube-attachment fittings arranged as illustrated for the flow-directing unit **104***a* in FIG. **1**) is illustrated in FIGS. **2**A-**2**C. The flow-directing unit **104***b* is similarly configured except for being a mirror image of the flow-directing unit

104*a*. The flow-directing unit **104***a* includes a generally circular, pan-shaped porting housing **120** and a disc-shaped, flow-controlling insert member **122** that fits into the porting housing **120**. Both components **120** and **122** may be made from medical-grade plastic or other suitable materials.

The porting housing **120** has a circular bottom wall **124** and a ring-shaped sidewall **126**, with an open top. As best shown in FIG. **2**A, tube attachment fittings **110** (corresponding to the attachment fittings **110***a*-**110***f* in FIG. **1**) are integrally formed with and extend outwardly from the sidewall **126**, as does tube attachment fitting **111**. The tube attachment fittings **110**, **111** are hollow and cylindrical, with open outer ends **128**, **130**, respectively, and are open to the interior of the porting housing **120** via apertures **132**, **134**, respectively. The tube attachment fittings **110**, **111** may have features (not illustrated) such as hose barbs, luer locks, etc., to secure the various tubes to the tube attachment fittings **110**, **111**. As best illustrated in FIG. **2**C, the tube attachment fittings **110** and corresponding apertures **132** are offset in the axial direction relative to the tube attachment fitting **111** and aperture **134**, with the apertures **132** being located closer to the open top of the sidewall **126** and the aperture **134** being located closer to the bottom wall **124**. The reason for this offset will be explained below.

The insert member **122**, on the other hand, has a generally puck-shaped body **136** and a circular positioning lip or flange **138** that extends circumferentially around the upper edge of the insert member **122**. As shown in FIG. **2**C, the positioning flange **138** limits the extent to which the body **136** of the insert member can be inserted into the interior of the porting housing **120**.

Furthermore, the insert member **122** has a notch **140** formed at the edge **142** of the insert member **122**, where the peripheral surface **144** of the insert member **122** meets the bottom surface **146** of the insert member. The notch **140** forms a passageway that permits fluid to pass between the bottom surface **146** of the insert member **122** from a selected one of the tube attachment fittings **110** into a space **148** and to the attachment fitting **111**. One of the attachment fittings **110** is selected depending upon the rotational position of the insert member **122**.

As further illustrated in FIG. **2**C, the insert member **122** is not as thick as the interior of the port housing **120** is deep. As a result, a chamber **148** is formed between the bottom surface **146** of the insert member and the bottom wall **124** of the port housing, and the apertures **134** are aligned with and opens into the chamber **148**. On the other hand, the apertures **132** are vertically positioned along the sidewall **126** (i.e., in the thickness direction) such that they are blocked by the peripheral surface **144** of the insert member, unless the notch **140** is positioned in front of a given one of the apertures **132**. For an aperture **132** that has the notch **140** positioned in front of it, fluid is able to flow between that aperture **132** and the chamber **148**. Thus, more broadly speaking, a fluid-flow pathway will extend through the flow-directing unit **104***a* from the tube attachment fitting **110** and aperture **132** having the notch **140** positioned in front of it; through the notch **140** and chamber **148**; and through the aperture **134** and tube attachment fitting **111**. Furthermore, it will be appreciated that different apertures **132** will be opened and other apertures **132** closed depending on the rotational position of the insert member **122** within the porting housing **120**.

A drive-engagement feature **150** is provided at the upper surface **152** of the insert member **122**. For example, as illustrated, the drive-engagement feature **150** could be a plus sign-shaped feature that stands proud relative to the upper surface **152** of the insert member. Alternatively, the drive-engagement feature could be a slot-shaped or cross-shaped recess; a post; a divot; gear teeth extending radially from the edge of the positioning flange **138**; or any other feature that can be engaged by a driving mechanism (illustrated and described below) and used to rotate the insert member **122** to a desired angular position within the porting housing **120**.

Additionally, a position-indicating feature **154** may also be provided on the upper surface **152** of the insert member **122**. The position-indicating feature **154** could be encoder markings that are detected by an optical sensor (illustrated and described below). Alternatively, the position-indicating feature could be indexing slots; magnets; or any other feature that can be sensed by a sensor to determine the angular position of the insert member **122**.

Further still, a circumferential recess **156** is suitably formed in the peripheral surface **144** of the insert member **122**, just under the positioning flange **138**. A sealing member **158** such as an O-ring made from medical-grade material fits within the circumferential recess **156** and bears against the radially inner surface **160** of the sidewall **126** to seal the interior of the flow-directing unit **104**. Additionally, means to secure the insert member **122** within the porting housing **120** (not illustrated) may also be provided. Such means may include clamps; a circumferential flange extending from the peripheral surface **144** of the insert member that engages with a corresponding circumferential groove formed in the radially inner surface **160** of the sidewall **126**; etc.

Use of the porting cassette **102** is illustrated in FIG. **3**. As shown in FIG. **1**, one of the cassette side parts **106***b* includes a pair of apertures **162***a*, **162***b* through which the flow-directing units **104***a*, **104***b* can be accessed. The porting cassette is inserted into a receiving slot **164** within the flow-porting section of an automated peritoneal dialysis system (not illustrated), with the flow-directing units **104***a*, **104***b* aligned with and accessible to reciprocating drive members **166***a*, **166***b* through the apertures **162***a*, **162***b*. The reciprocating drive members **166***a*, **166***b* have features (e.g., end faces) that are configured to engage with the drive-engagement features **150** of the flow-distribution units **104***a*, **104***b* so that the drive members **166***a*, **166***b* can rotate the insert members **122** of the flow-distribution units **104***a*, **104***b* to various positions to open and close the various apertures **132** of the flow-distribution units. By adjusting the rotational positions of the insert members **122** of the flow-distribution units **104***a*, **104***b*, the flow of fluid through the incoming-fluid delivery tubes **108***a*-**108***f* and the outgoing-fluid delivery tubes **112***a*-**112***f* can be regulated.

The flow-porting section of the automated peritoneal dialysis system includes a controller **170**, which receives a flow path command from the system to establish a desired combination of incoming fluid path and outgoing fluid path. The controller **170** then commands stepper motors **172***a*, **172***b* to drive the drive members **166***a*, **166***b* to commanded angular positions to achieve the desired flow path. Furthermore, position sensors **174***a*, **174***b* detect the position-indicating features **154** on the flow-distribution units **104***a*, **104***b*. In this manner, the controller **170** is provided with the necessary information to control the positions of the insert members **122** of the flow-distribution units **104***a*, **104***b* and hence to control the overall fluid-flow pathway.

Given the relatively compact design of the fluid-distribution units, they can be fabricated relatively inexpensively. This is beneficial for medical components that are to be disposed of. Additionally, the design reduces complexity of the overall peritoneal dialysis system in that the settings for just two components—namely, the angular positions of the insert members of the two flow-distribution units—needs to be regulated instead of the actuation states of clamping devices on each of the various fluid-flow lines. Further still, the design affords high assurance that flow will be prevented or allowed through the various lines.

Another configuration of a flow-directing unit **200** is illustrated in FIGS. **4**A and **4**B. Each flow-directing unit **200** has a flow barrel **214** with ports **206** connecting fluid circuit elements, one of which is indicated at **204**, to form a manifold. Each flow-directing unit **200** has a selector drum **208** that engages with a selector drive (not shown but similar to that shown in FIG. **3**) by means of a drive-engagement feature **212**. Ports **206** align with the ports **210** to connect a selected port **206** to a common port **220**. The flow directing unit **200** is used in symmetrical pairs as are flow-directing units **104***a* and **104***b*. Each flow-directing unit **200** common port **220** connects to a pumping tube segment (not shown but arranged similarly to the arrangement shown in FIG. **1**). Thus each of two of the flow-directing units **200** would have its port **220** facing the other and connected to that other port **220** by the pumping tube segment (again, not shown). The selector drum **208** is shown extended out of the flow barrel **214** to reveal the ports **210** that would otherwise be hidden, but it will be understood that the selector drum is positioned inside the barrel **214** during use. Each port **210** may have an O-ring **211** to seal it against the inner wall of the barrel **214** to form a seal with a respective one of the ports **206**. It will be evident from the drawing and the above description that independent pump inlet and pump outlet channels of a fluid circuit may be selected using the above configuration. Although not shown, indicia can be provided on the ends of the selector drum **208** to indicate the angular position of the selector drum.

According to embodiments, the disclosed subject matter includes a disposable medical flow-regulating device having a pair of cylindrical flow-directing units, with each of the flow-directing units having a housing with tube attachment fittings directed approximately radially away from an axis of the flow-directing unit and a transfer fitting. A pumping tube segment extends from the transfer fitting on one of the flow-directing units to the transfer fitting on the other of the flow-directing units to establish a fluid flow path between the two flow-directing units. Each of the flow-directing units having a disc-shaped, rotary insert member rotates within a chamber to select one of the tube attachment fittings at a given angular position thereof to connect with respective one of the transfer fittings whereby a selectable channel from a first flow-directing unit tube attachment fitting to a second flow-directing unit tube attachment fitting is defined. Tubing elements from a fluid circuit are connected the tube attachment fittings of each of the flow-directing units such that selectable flow paths in the fluid circuit may be defined by rotating the rotary insert members.

In further variations of the embodiments, each of the rotary insert members seals to a respective housing using an O-ring. In further variations of the embodiments, a flow chamber is defined between each rotary insert member and a respective one of the housings. In further variations of the embodiments, there may be included a respective rotary actuator that engages with a respective one of the rotary insert members. In further variations of the embodiments, the pair of cylindrical flow-directing units and the pumping tube segment are partially enclosed in a support member to form a cartridge enclosure. In further variations of the embodiments, the support has openings to provide access to the rotary insert members and the pumping tube segment.

According to further embodiments, the disclosed subject matter includes a selector valve with first and second flow switches, each having a cylindrical chamber with a rotary element that selectively interconnects a common port with a selected one of a plurality of individual ports. The cylindrical chamber and rotary element forming a fluid passage defined by a hollow space between them. The common port of each of said pair being connected to a respective end of a pumping tube segment. A rotary actuator is provided for each of said rotary elements and a controller configured to rotate the rotary elements independently to define selected interconnections in a fluid circuit connected to the plurality of individual ports.

In variations thereof, the embodiments include ones in which each of the rotary elements seals to a respective one of the cylindrical chambers by means of an O-ring.

In variations thereof, the embodiments include ones in which the pair of flow switches and the pumping tube segment are partially enclosed in a support member to form a cartridge enclosure. In variations thereof, the embodiments include ones in which the support has openings to provide access to the rotary elements and the pumping tube segment.

We claim:

1. A disposable medical flow-regulating device, comprising:

a first cylindrical flow-director and a second cylindrical flow-director, each of the first and second cylindrical flow-directors including

a pan-shaped hollow housing having a circular bottom wall and a side wall around an outer perimeter of the circular bottom wall, with a plurality of tube attachment fittings extending out of an outer surface of the side wall, each of the tube attachment fittings being in fluid communication with an interior cavity of the pan-shaped hollow housing,

a first tube attachment fitting of the plurality of tube attachment fittings having a position closer to the circular bottom wall than all others of the plurality of the tube attachment fittings, and

a rotary insert that is configured to rotate within the interior cavity of the pan-shaped hollow housing, the rotary insert having a peripheral wall with a height that is less than a height of the side wall of the pan-shaped hollow housing, such that the peripheral wall occludes the others of the plurality of the tube attachment fittings but does not occlude the first tube attachment fitting when the rotary insert is fully inserted into the pan-shaped hollow housing, and the rotary insert having a notch in the peripheral wall that allows a selected one of the others of the plurality of the tube attachment fittings to fluidly communicate with the first tube attachment fitting when the notch is positioned adjacent to the selected one of the others of the plurality of the tube attachment fittings;

a pumping tube segment compatible with a peristaltic pump, the pumping tube segment extending from the first tube attachment fitting of the first cylindrical flow-director to a first tube attachment fitting of the second cylindrical flow-director to establish a fluid flow path between the first and the second cylindrical flow-directors;

fluid delivery tubes connected to the others of the plurality of the tube attachment fittings of each of the first and second cylindrical flow-directors such that selectable flow paths between the fluid delivery tubes are defined by rotating a respective rotary insert.

2. The device of claim **1**, wherein each rotary insert seals to a respective pan-shaped hollow housing using an O-ring.

3. The device of claim 1, wherein a flow chamber is defined between each rotary insert and the circular bottom wall of a respective pan-shaped hollow housing.

4. The device of claim 1, further comprising a drive-engagement feature on a top surface of each rotary insert that is configured to engage with a driving mechanism.

5. The device of claim 1, wherein each of the first and second cylindrical flow-directors and the pumping tube segment are partially enclosed in a support frame to form a cartridge enclosure.

6. The device of claim 5, wherein the support frame has openings to provide access to the rotary inserts and the pumping tube segment.

7. The device of claim 1, wherein the rotary inserts are disc shaped.

* * * * *